1. Create a class named 'Student' with data members: string 'name', integer 'rollNo', string 'phoneNo' and string 'address'. Also add static member to count number of students. Write following member function:

- non-parameterized constructor. Assign 1 as roll no. Assign empty string to remaining members
- parameterized constructor with single parameter roll no. Assign empty string to remaining members
- set function with single parameter roll no. Check roll no (should be non-zero), otherwise assign 1 as roll no. Assign empty string to remaining members
- overload stream insertion operator (see example output below)
- write getter to get number of students
- write appropriate destructor
- do appropriate handling of static member count

In main create object of class Student using single parameterized constructor. Print object using cout. Next ask user to enter a number. Create dynamic array of students according to number. Call setter for array and print values using cout. Print number of students. Delete dynamic array of students. Create a dynamic single object. Call setter for dynamic object. Again print number of students. Finally delete dynamic object.

Note: Avoid duplication of code to get maximum numbers.

Example Output:

Name: Ali Akbar

Roll No: BSEF01M501

Phone No: 03332233445

Address: 255 Gulshan e Iqbal Karachi

2. Write a program to print the area and perimeter of a rectangle. Create class Rectangle with set, calculateArea, calculatePerimeter member functions. In main declare an object and set values of length and width. Print area and perimeter by calling respective member functions.

3. Create a class BinaryCalculator. Class has two data members (two numbers). Create member functions to calculate & return sum, difference, product of two numbers. Write main to test your class.